INACTIVATED FOR NEW DESIGN
AS OF 10 JULY 1995

INCH-POUND

MIL-M-38510/116
AMENDMENT 6
14 MAR 1996
SUPERSEDING
AMENDMENT 5
20 May 1983

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, CMOS, ANALOG SWITCH WITH DRIVER,

MONOLITHIC SILICON

This amendment forms a part of Military Specification *MIL-M-38510/116*, dated 16 April 1980, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

TITLE, delete and substitute new title as printed above.

1.3, V_{IN} to ground: Delete "V^--36 Vdc" and substitute "V^+-36 Vdc".

* 1.3, $+V_{CC}$ to $-V_{CC}$: Delete "+36 Vdc" and substitute "44 Vdc".

PAGE 2

1.4, last line: Delete "Case" (as substituted by Amendment 2) and substitute "Ambient"; delete "T_C" (as substituted by Amendment 2) and substitute "T_A".

1.5, Maximum allowable power dissipation column, Case outline I: Delete "250 mW" and substitute "350 mW"; delete "T_C" (3 places) (as substituted by Amendment 2) and substitute "T_A".

3.4, line 2: Delete "case" (as substituted by Amendment 2) and substitute "ambient".

PAGE 3

TABLE I, R_{DS}, Conditions column, first listing: Delete "V_D = 10 V" and substitute "V_D = -10 V".

TABLE I, $I_{D(ON)}$, Limits column: Delete "-200, 200, -2, 2, -100, 100, -1, 1" and substitute "-100, 100, -1, 1, -200, 200, -2, 2", respectively.

TABLE I, I_{IH}, Device type column: Delete "01, 02, 03, 09" and substitute "01, 02, 03, 04".

TABLE I, Temperature column: Delete "T_C" (all locations) (as substituted by Amendment 2) and substitute "T_A".

PAGE 4

TABLE I, Temperature column: Delete "T_C" (all locations) (as substituted by Amendment 2) and substitute "T_A".

TABLE I, *Break-before-make time delay test*, Min limits column: Delete "20" and substitute "5".

TABLE I, driver input capacitance tests, maximum limits column: Delete "6" and "3.5" and substitute "15" and "10" respectively.

TABLE I, switch input capacitance test, maximum limits column: Delete "14" and substitute "30".

TABLE I, Switch output capacitance test, maximum limits column: Delete "14" and substitute "30".

TABLE I, input test voltage (V_{ZAP}): Delete in it's entirety.

FSC 5962

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TABLE I, footnote 1/, delete and substitute:

"1/ The listed resistance limits correspond to the following voltage values:

9.5 V,	-9.5 V	50Ω	6.8 V,	-6.8 V	70Ω
9.25 V,	-9.25 V	75Ω	6.5 V,	-6.5 V	100Ω

3.5, delete and substitute the following:

"3.5 Electrical test requirements. Electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III."

TABLE II, delete and substitute:

"TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical test parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*,2,3,9	1*2,3,9	1
Group A test requirements (method 5005)	1,2,3,9,10, 11,12***,13, (4,7)****	1,2,3,9,10, 11,12***,13 (4,7)****	1,2,3,9, (4,7)****
Group C end-point and group B class S electrical parameters (method 5005)	1,2,3, and table IV delta limits	1 and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups for group C periodic inspections	N/A	(4,7)*****	10,11,12,13 (4,7)*****
Group D end-point electrical parameters (method 5005)	1,2,3	1	1
Additional electrical subgroups for group D periodic inspections	(4,7)**	None	None

* PDA applied to subgroup 1 (see 4.3e).
** See 4.4.4b.
*** See 4.4.1d.
**** See 4.4.1e.
***** See 4.4.3c."

3.7, line 2: Delete "57" and substitute "82".

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PAGE 10

FIGURE 4: Delete schematics for device types 01 and 02 and substitute new schematics as shown on pages 9 and 10 of this amendment. (Device 02 schematic reprinted without change from amendment 4).

PAGE 11

FIGURE 4: Delete schematic for device type 03 and substitute new schematic as shown on page 10 of this amendment.

PAGE 12

FIGURE 4: Delete schematic for device type 04 and substitute new schematic as shown on page 11 of this amendment.

PAGE 13

FIGURE 4: Delete schematics for device types 05 and 06 and substitute new schematics as shown on page 12 of this amendment.

PAGE 14

FIGURE 4: Delete schematic for device type 07 and substitute new schematic as shown on page 13 of this amendment.

PAGE 15

FIGURE 4: Delete schematic for device type 08 and substitute new schematic as shown on page 14 of this amendment.

PAGE 16

FIGURE 5, right hand illustration, at the current source below +V: Delete "I_S = -10 mA" and substitute "I_S = 10 mA".

PAGE 19

* FIGURE 11: Note 1b, for part types 01 - 04 rise and fall times, change from "≤ 10 ns" to "≤ 20 ns".

PAGE 20

FIGURE 14: Delete and substitute new figure as shown on page 15 of this amendment.

PAGE 21

* FIGURE 15: Change V_X from "8 V" to "9 V" for condition A, from "-8 V" to "-9 V" for condition B. Note 4, change V_X from "8 V" to "9 V" for condition A, from "-8 V" to "-9 V" for condition B. Note 5b, for part types 01 - 04 rise and fall times, change from "≤ 10 ns" to "≤ 20 ns".

PAGE 31

TABLE III, Subgroup column: Delete "T_C" (all locations) (as substituted by Amendment 2) and substitute "T_A"; make the same changes on pages 32 through 60.

PAGE 32

TABLE III, test 93, adapter pin no 1 column: Delete "0.8V" and substitute "4.0V";
adapter pin no 14 column: Delete "2.0V" and substitute "0.8"; test 94,
adapter pin no 1 column: Delete "4.0V" and substitute "0.8V";
test 95, adapter pin no 14 column: Delete "4.0V" and substitute "0.8V".

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PAGE 33

TABLE III, tests 108 and 109, maximum limits column: Delete "6" and substitute "15";
tests 110 and 111, maximum limits column: Delete "3.5" and substitute "10";
tests 112, 113, 114, 115, maximum limits column: Delete "14" and substitute "30".

PAGE 35

TABLE III, test 51, Max limits column: Delete "0.01" and substitute "0.1";
test 53, Min limits column, delete "-0.01" and substitute "-0.1";
test 54, Min limits column, delete "-0.01" and substitute "-0.1".

PAGE 36

TABLE III, test 99, maximum limits column: Delete "6" and substitute "15";
test 100, maximum limits column: Delete "3.5" and substitute "10";
tests 101, 102, 103, 104, maximum limits column: Delete "14" and substitute "30";

* TABLE III, tests 105 and 106, Min limits column: Delete "5" (as substituted by amendment 4) and substitute "15". Make this same change on page 45, tests 197-200; page 51, tests 102 and 103; page 60, tests 193-196.

PAGE 40

TABLE III, tests 185 and 186, maximum limits column: Delete "6" and substitute "15";
tests 187 and 188, maximum limits column: Delete "3.5" and substitute "10";
tests 189 thru 196, maximum limits column: Delete "14" and substitute "30".

TABLE III, test 155, adapter pin no 14 column: Add "0.8V";
test 156, adapter pin no 1 column: Add "0.8V".

PAGE 44

TABLE III, test 155, measured pin number column: Delete "12" and substitute "11";
test 156, adapter pin number 3 column: Delete "IN 4/";
test 156, adapter pin number 4 column: Add "IN 4/";
test 156, measured pin number column: Delete "10" and substitute "13";
test 158, adapter pin number 1 column: Delete "0.8V" and substitute "4.0V";
test 160, adapter pin number 14 column: Delete "0.8V" and substitute "4.0V".

TABLE III, tests 185 and 186, maximum limits column: Delete "6" and substitute "15";
tests 187 and 188, maximum limits column: Delete "3.5" and substitute "10";
tests 189 thru 198, maximum limits column: Delete "14" and substitute "30".

PAGE 46

* TABLE III, tests 5-8, 33-36, change columns 1 and 14 from "11.0 V" to "7.0 V".

PAGE 47

* TABLE III, tests 61-64, change columns 1 and 14 from "11.0 V" to "7.0 V".

TABLE III, test 87, adapter pin number 1 column: Delete "3.5V" and substitute "11.0V";
test 87, adapter pin number 14 column: Delete "11.0V" and substitute "3.5V";
test 88, adapter pin number 1 column: Delete "11.0V" and substitute "3.5V";
test 89, adapter pin number 14 column: Add "3.5V".

PAGE 48

TABLE III, tests 103 and 102, maximum limits column: Delete "6" and substitute "15";
tests 104 and 105, maximum limits column: Delete "3.5" and substitute "10";
tests 106 thru 109, maximum limits column: Delete "14" and substitute "30".

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PAGE 49

* TABLE III, tests 5, 7, 31, 33, change column 1 from "3.5 V" to "2.5 V".

* TABLE III, tests 6, 8, 32, 34, change column 1 from "11.0 V" to "7.0 V".

PAGE 50

* TABLE III, tests 57 and 59, change column 1 from "3.5 V" to "2.5 V".

* TABLE III, tests 58 and 60, change column 1 from "11.0 V" to "7.0 V".

PAGE 51

TABLE III, test 96, maximum limit column: Delete "6" and substitute "15";
test 97, maximum limit column: Delete "3.5" and substitute "10";
tests 98 thru 101, maximum limit column: Delete "14" and substitute "30".

PAGE 52

* TABLE III, tests 9-16, change column 1 and 14 from "11.0 V" to "7.0 V".

PAGE 53

* TABLE III, tests 57-64, change column 1 and 14 from "11.0 V" to "7.0 V".

PAGE 54

* TABLE III, tests 105-112, change column 1 and 14 from "11.0 V" to "7.0 V".

PAGE 55

TABLE III, test 149, adapter pin number column: Add "3.5V";
test 149, measured pin number column: Delete "11" and substitute "12";
test 150, adapter pin number 1 column: Delete "3.5V" and substitute "11.0V";
test 150, adapter pin number 14 column: Add "3.5V";
test 150, measured pin number column: Delete "10" and substitute "13";
test 151, adapter pin number 1 column: Add "3.5V";
test 151, measured pin number column: Delete "13" and substitute "10";
test 152, adapter pin number 1 column: Add "3.5V";
test 152, adapter pin number 14 column: Delete "3.5V" and substitute "11.0V";
test 152, measured pin number column: Delete "12" and substitute "11";
test 154, adapter pin number 1 column: Delete "11.0V" and substitute "3.5V";
test 156, adapter pin number 14 column: Delete "11.0V" and substitute "3.5V";
tests 181, 182, maximum limit column: Delete "6" and substitute "15";
tests 183, 184, maximum limit column: Delete "3.5" and substitute "10";
tests 185 thru 192, maximum limit column: Delete "14" and substitute "30".

PAGE 56

* TABLE III, tests 9, 11, 13, and 15, change columns 1 and 14 from "11.0 V" to "7.0 V".

* TABLE III, tests 10, 12, 14, and 16, change columns 1 and 14 from "3.5 V" to "2.5 V".

PAGE 57

* TABLE III, tests 57, 59, 61, and 63, change columns 1 and 14 from "11.0 V" to "7.0 V".

* TABLE III, tests 58, 60, 62, and 64, change columns 1 and 14 from "3.5 V" to "2.5 V".

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PAGE 58

* TABLE III, tests 105, 107, 109, and 111, change columns 1 and 14 from "11.0 V" to "7.0 V".

* TABLE III, tests 106, 108, 110, and 112, change columns 1 and 14 from "3.5 V" to "2.5 V".

PAGE 59

TABLE III, tests 181, 182, maximum limit column: Delete "6" and substitute "15";
tests 183, 184, maximum limit column: Delete "3.5" and substitute "10";
tests 185 thru 192, maximum limit column: Delete "14" and substitute "30".

PAGE 60

TABLE III, tests 193-196: Adapter pin number 6, add "15 V"; adapter pin number 7, add "-15 V"; adapter pin number 8, add "GND".

TABLE III, note 3, line 3: Delete "duty cycle." and substitute "duty cycle, or single step changing device from "ON" state to "OFF" state (rise time/fall time $\leq$ 20nS)."

PAGE 61

4.3a, add the following: "NOTE: Burn-in circuit resistor tolerances for figures 16, 17 and 18 shall be ±10%."

4.3a, item 1, line 1: Delete "(test condition A)" and substitute "(test condition C)".

4.3b, delete and substitute the following:

"b. Reverse bias burn-in (method 1015 of MIL-STD-883) required for class S devices only, using the circuit shown on figure 16. However, regardless of device class, for devices 02, 04, 06, and 08, an additional burn-in shall be performed with the logic level of the switch drivers opposite that used in the first burn-in. Ambient temperature (T_A) shall be 125°C minimum. Duration for the reverse bias test (class S devices) shall be 24 hours minimum. Duration for additional burn-in (class B devices) shall be 160 hours minimum."

4.4.1c, delete and substitute:

"c. Subgroups 12 and 13 shall be added to group A inspection as specified in table III herein. For subgroup 12, device groups (01, 02, 05, 06) and (03, 04, 07, 08) may be qualified by passing qualification tests on the first device type from each group that is submitted to qualification. Subgroup 12 shall be performed using a sample of 5 devices with no failures allowed. The LTPD for subgroup 13 shall be 10 for all classes."

4.4.1, delete subparagraph e (as added in Amendment 3) and substitute:

"e. Subgroups 4 and 7 shall be performed for initial qualification only using a sample of 5 devices for each device type submitted to group A inspection, with no failures allowed. If not more than 1 failure is found in the first sample of 5, a second sample of 5 is permitted with no further failures allowed."

PAGE 62

4.4.2b, add the following sentence: "For device types 02, 04, 06, and 08, life test duration shall be divided equally between forward bias and reverse bias."

4.4.2c: Delete in it's entirety.

4.4.3b, add the following sentence: "For device types 02, 04, 06, and 08, life test duration shall be divided equally between forward bias and reverse bias."

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4.4.3, add new subparagraph c as follows:

"c. A special subgroup shall be added to group C inspection for classes B and C devices only and it shall consist of the group A subgroups 4 and 7 as specified in table III herein. This special subgroup shall be performed on each device type that is qualified from those listed in 1.2.1 herein. After initial qualification, the special subgroup shall be performed periodically on a single device type selected from those device types previously qualified. A sample of 5 devices (of the device type to be inspected) shall be chosen and submitted to test with no failures allowed. If not more than 1 failure is found in the first sample of 5, a second sample of 5 is permitted with no further failures allowed. When more than one device type is qualified, the single device type selected shall be a different device type for each subsequent periodic inspection until all qualified device types have been inspected. The sequence of single device types shall be repeated to fulfull the periodic inspection requirement."

4.4.4b, delete (as substituted by Amendment 3) and substitute:

"b. A special subgroup shall be added to group D inspection for class S devices only and it shall consist of the group A subgroups 4 and 7 as specified in table III herein. This special subgroup shall be performed on each device type that is qualified from those listed in 1.2.1 herein. After initial qualification, the special subgroup shall be performed periodically on a single device type selected from those device types previously qualified. When more than one device type is qualified, the single device type selected shall be a different device type for each subsequent periodic inspection until all qualified device types have been inspected. The sequence of single device types shall be repeated to fulfill the periodic inspection requirement."

4.4.4c, delete in it's entirety.

TABLE IV, under the title: Delete "T_C" (as substituted by Amendment 2) and substitute "T_A".

PAGE 63

4.5.3: Delete in it's entirety.

PAGE 65

Custodians column: Add "NASA-NA".

Review activities column, delete and substitute the following:

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99

User activities column, delete and substitute the following:

User activities:
Army - SM
Navy - AS, CG, MC

* User activities column, delete in its entirety.

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Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99

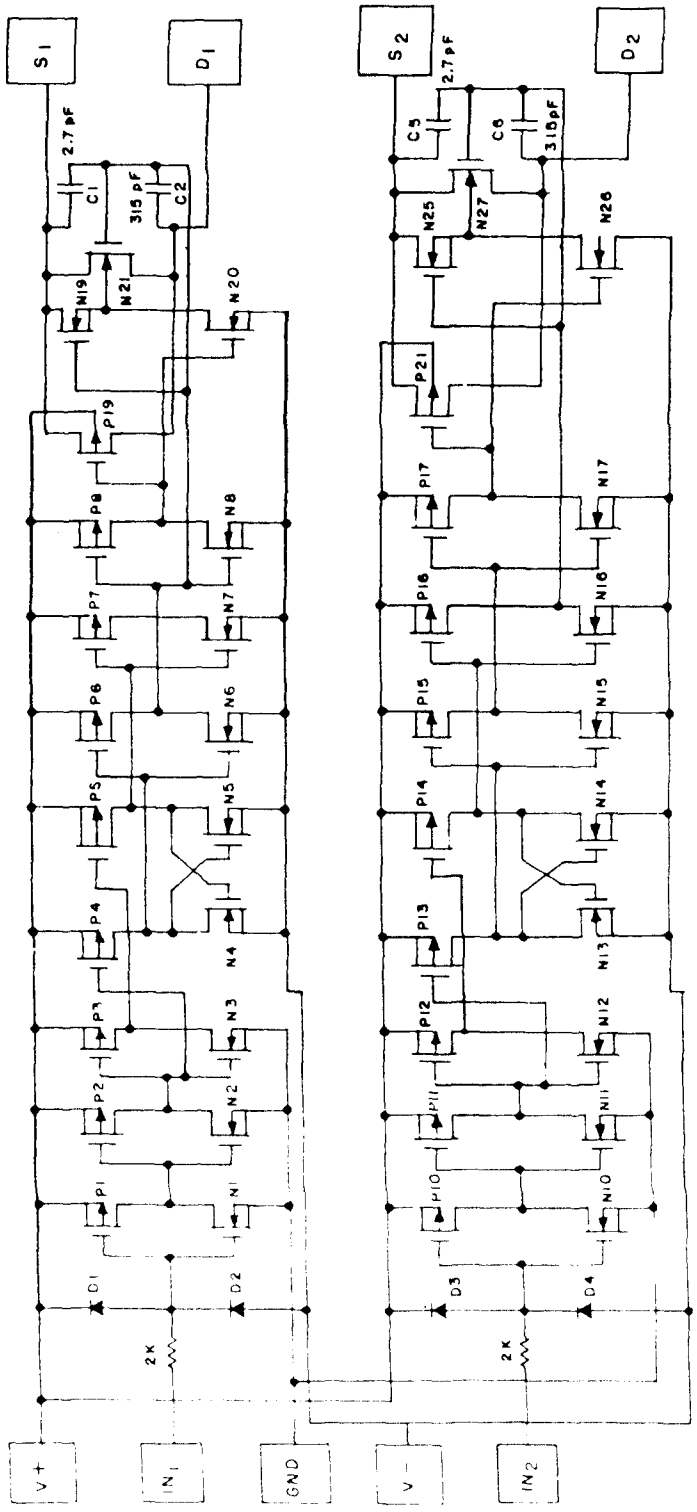
Agent:
DLA - ES

Preparing activity:
DLA - ES

(Project 5962-1550)

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Device type 01

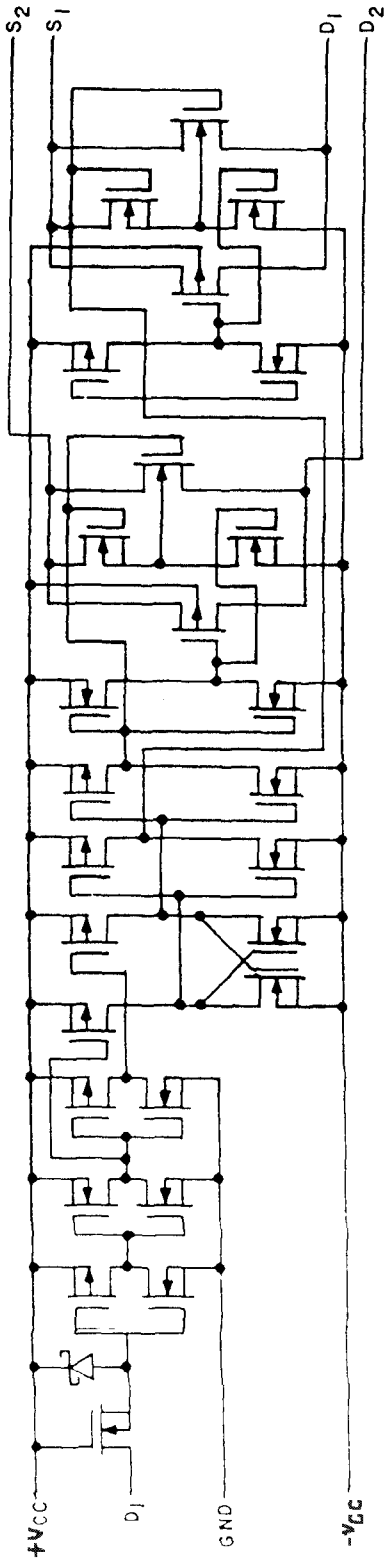


SILICONIX, INC.

FIGURE 4. Schematic circuit.

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Device type 02



SILICONIX, INC.

FIGURE 4. Schematic circuit - Continued.

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Device type 03

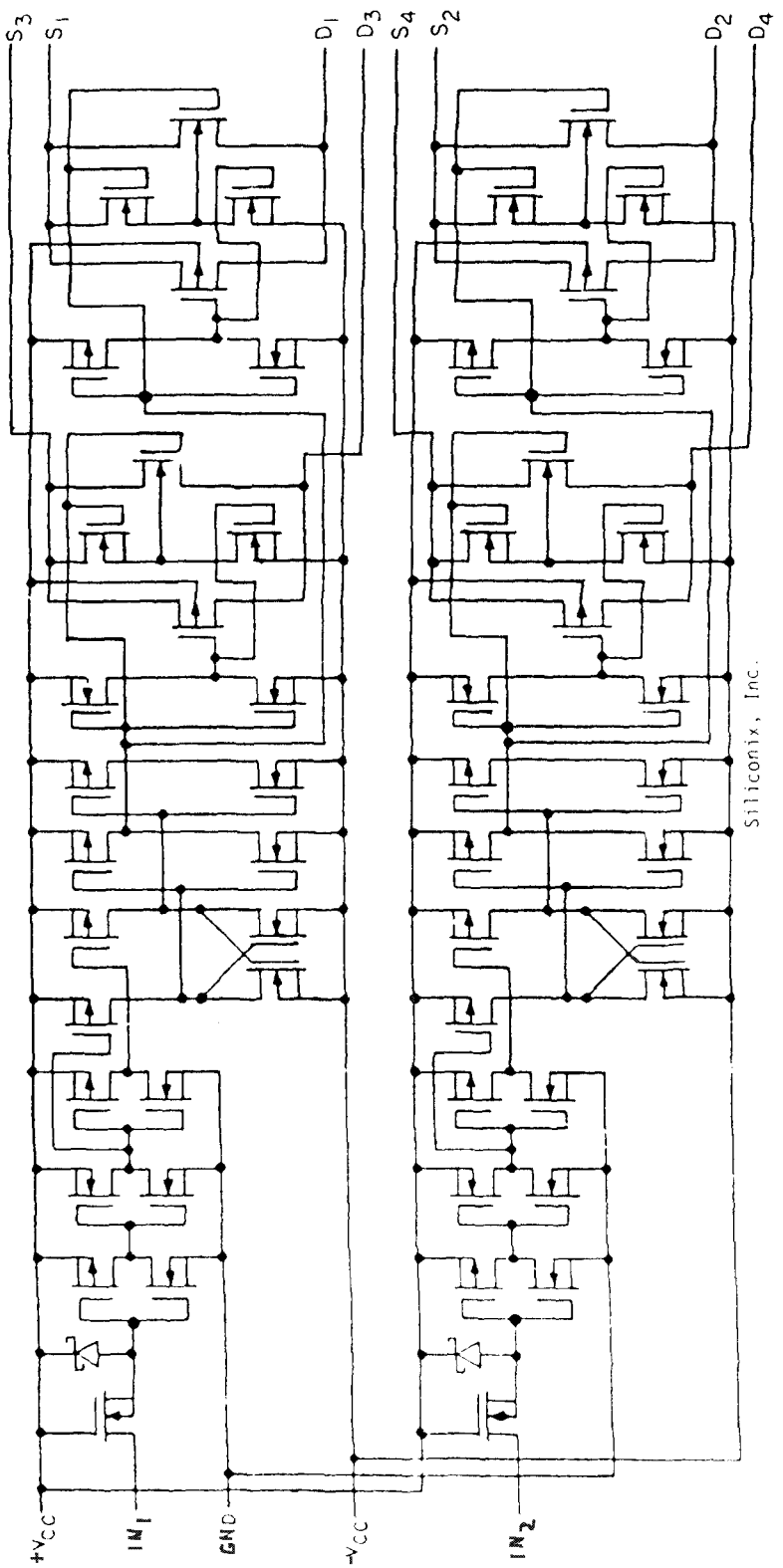


FIGURE 4. Schematic circuit - Continued.

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Device type 04

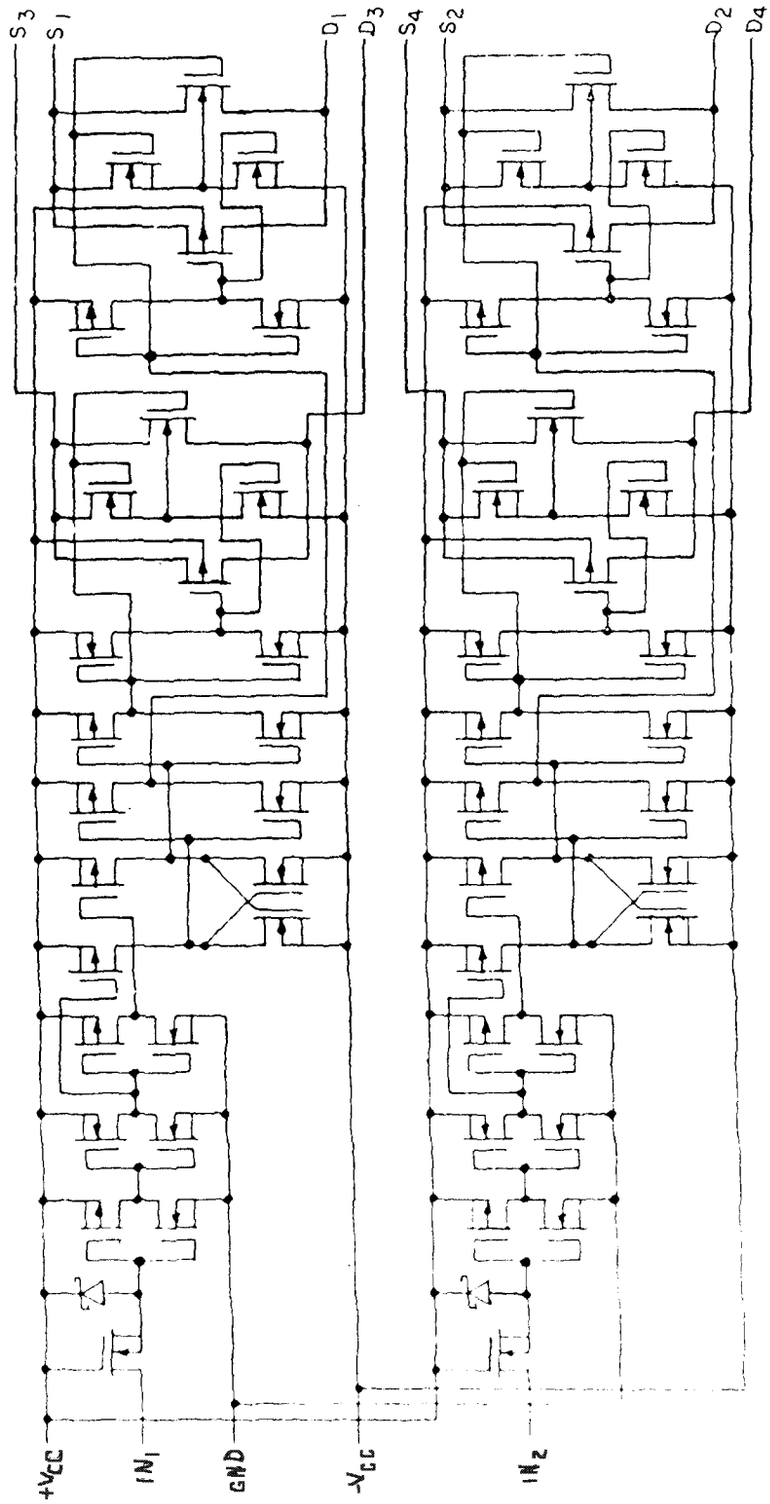


FIGURE 4. Schematic circuit - Continued.

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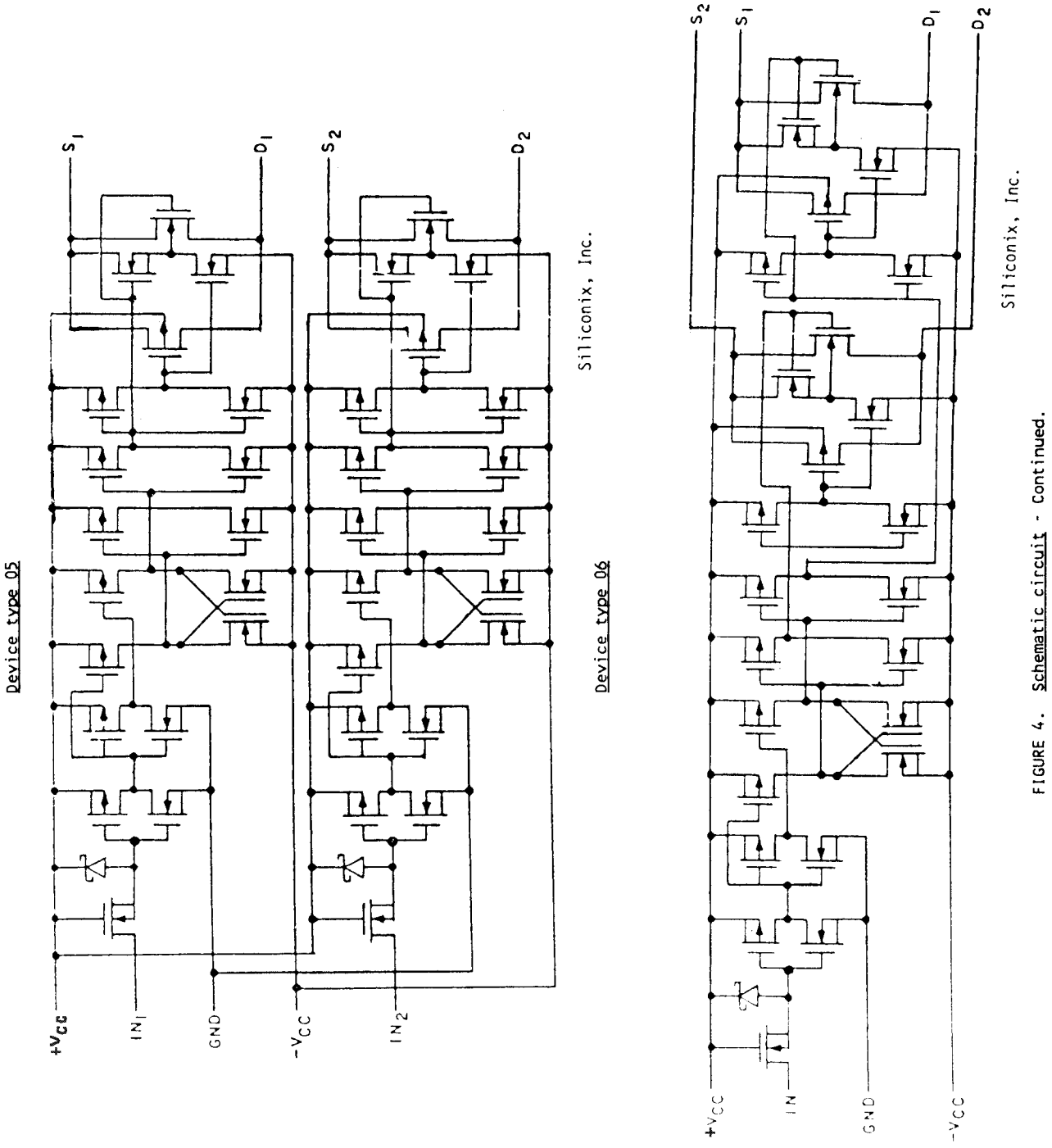


FIGURE 4. Schematic circuit - Continued.

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Device type 07

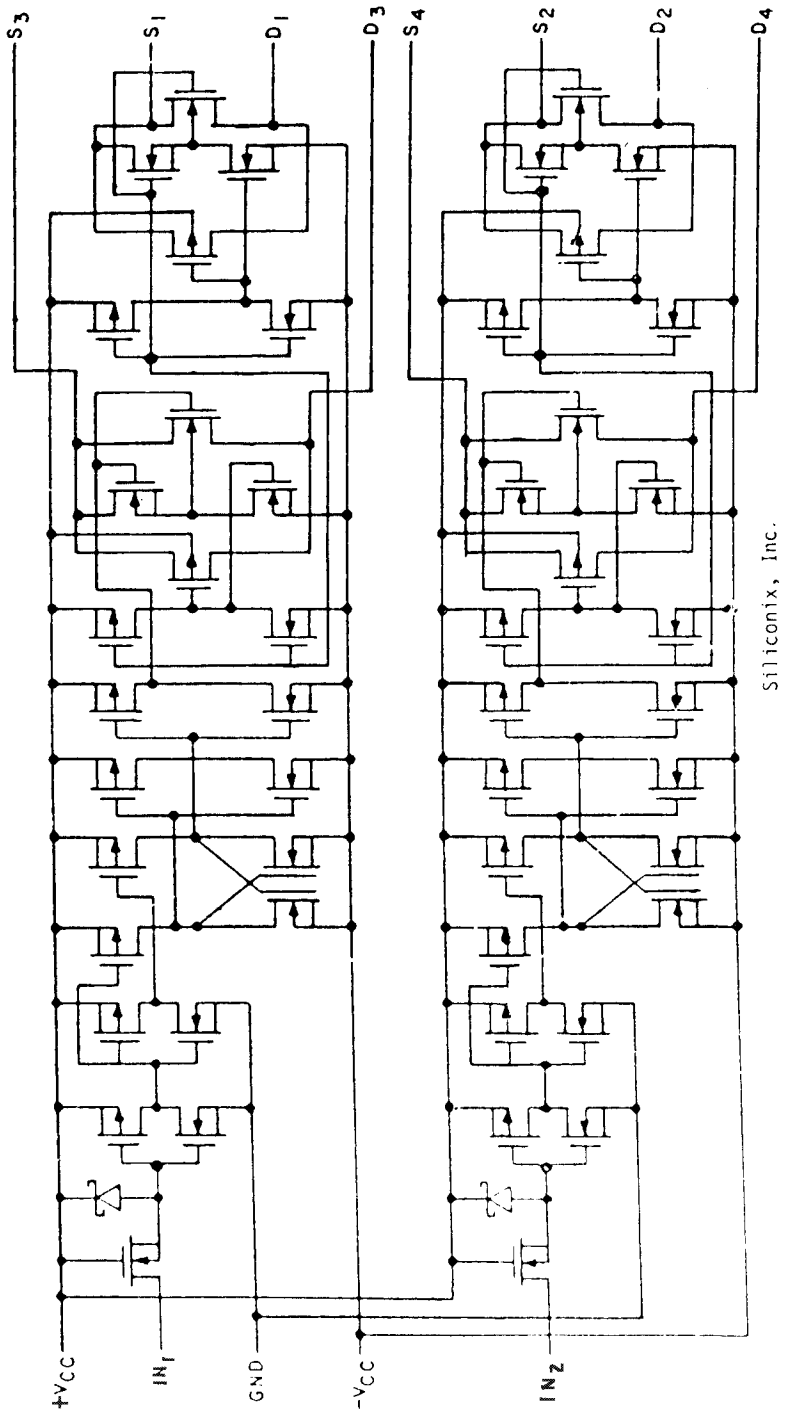


FIGURE 4. Schematic circuit - Continued.

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Device type 08

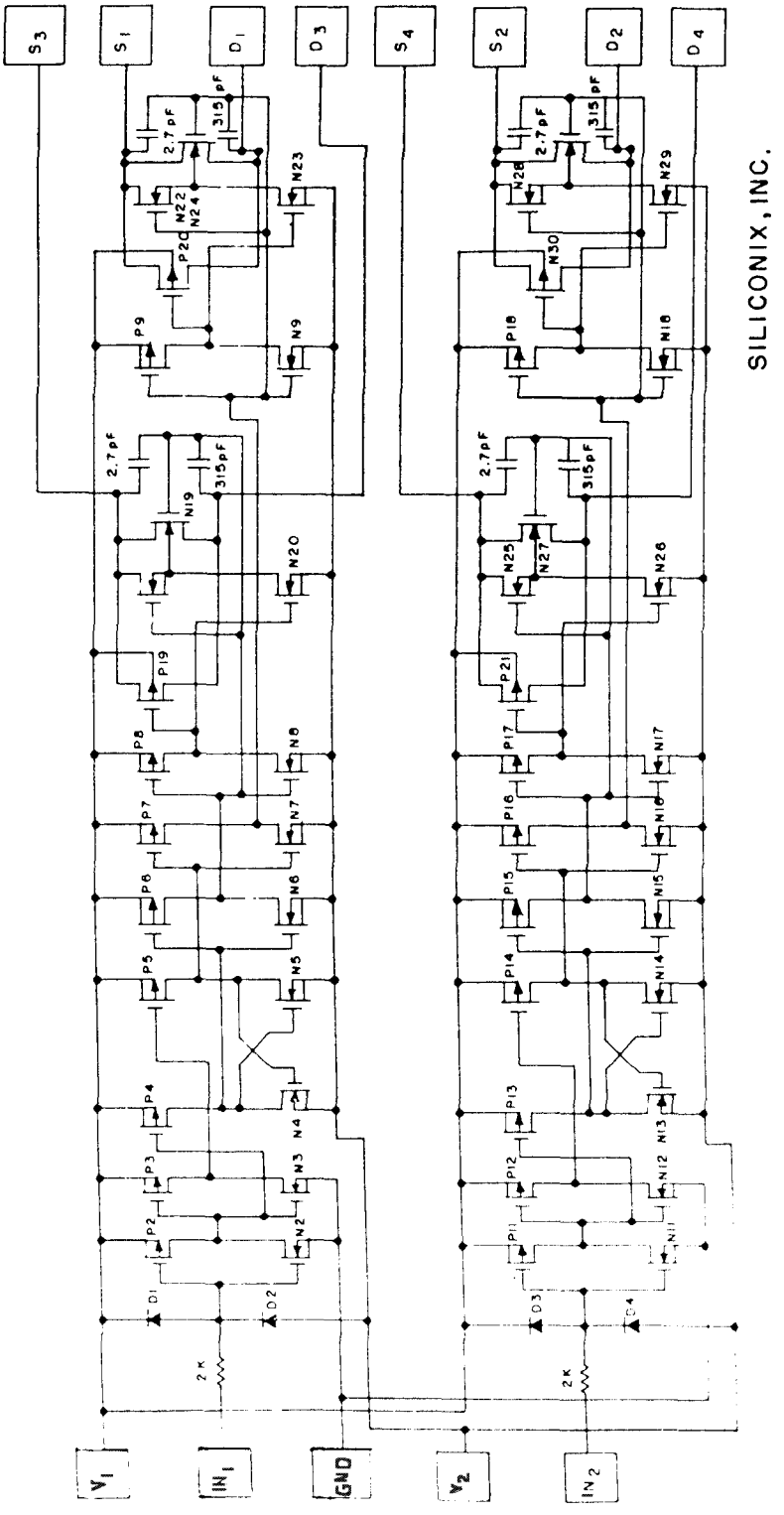


FIGURE 4. Schematic circuit - Continued.

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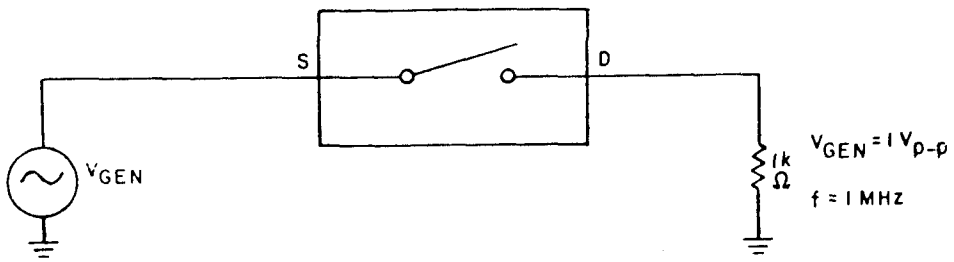


FIGURE 12. Isolation test circuit.

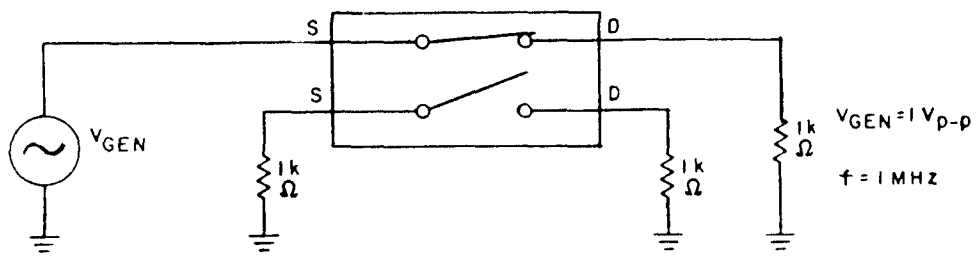
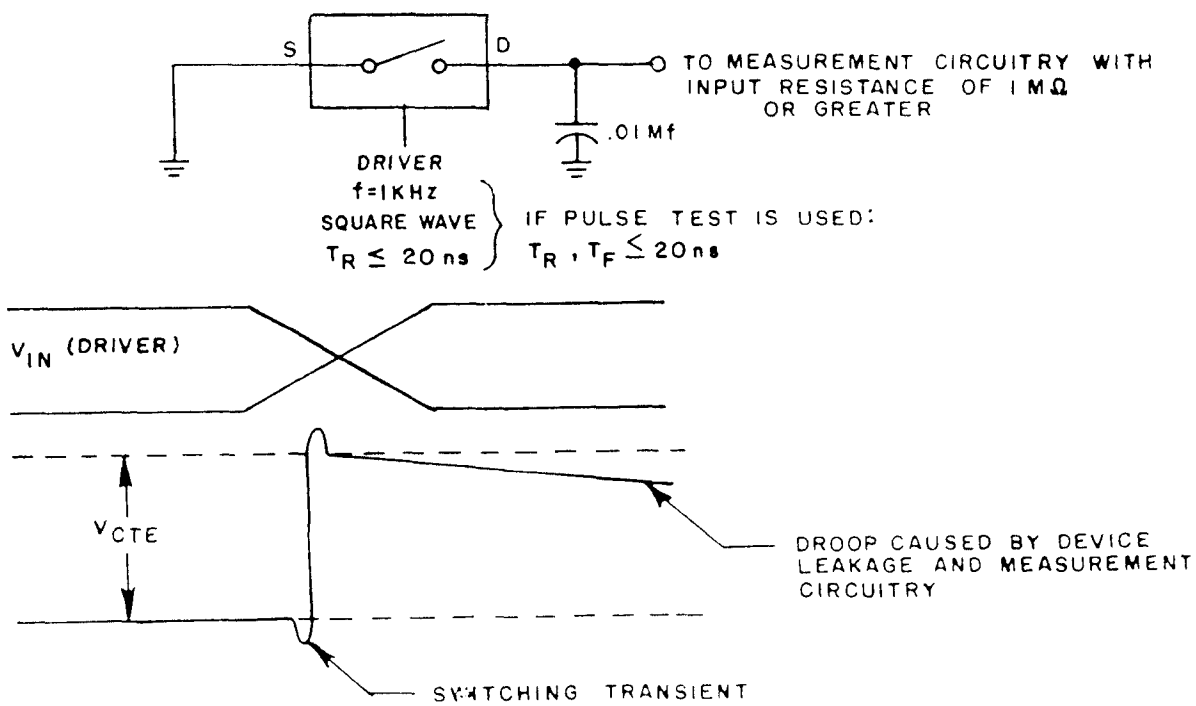


FIGURE 13. Crosstalk test circuit.



NOTE: V_{CTE} may be a positive or negative value.

FIGURE 14. Charge transfer error test circuit.

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